POPURI HEMANTH KUMAR

+91 9959942133 popuri.hemanthkumar@gmail.com Hyderabad
www.linkedin.com/in/hemanth-popuri https://github.com/Popuri-Hemanth
https://hemanth-portfolio-6u45.vercel.app/

ABOUT ME

SoftwareDeveloper with strong foundations in JavaScript, full-stack web development, and data structures & algorithms. Experienced in building modern web applications using React.js, Node.js, and REST APIs, with hands-on knowledge of database systems and authentication mechanisms. Skilled in developing scalable backend services, interactive user interfaces, and AI-assisted computer vision applications. Passionate about solving complex problems and building efficient software systems.

SKILLS

- **Programming Languages :** JavaScript, Python, C#, C
- **Frontend Technologies :** React.js, HTML5, CSS3
- **Backend Technologies :** Node.js, Express.js
- **.NET Technologies :** ASP.NET Core, .NET Runtime, C#, Entity Framework, LINQ
- **Databases :** MongoDB, PostgreSQL
- **Computer Vision Tools :** OpenCV, Image Processing, Object Detection
- **AI / Dataset Platforms :** Kaggle, Roboflow
- **Deployment Platforms :** Railway
- **Version Control & Development Tools :** Git, GitHub, VS Code, Postman
- **Community & Research Platforms :** Reddit
- **Operating Systems :** Linux, Windows

PROJECTS

AI-PoweredAutomated Rummy Table Scoring System

- Developed an AI-based automated scoring system for Rummy clubs that detects playing cards from table images and calculates player scores automatically using computer vision and deep learning.
- Implemented a YOLO-based object detection pipeline to identify playing cards, player badge numbers, and DROP labels from a captured table image.
- Designed a rule-based scoring engine supporting both 13-card and 21-card Rummy variants, automatically assigning cards to players and calculating penalties or winning scores.
- Integrated a camera-triggered image capture system with dataset preprocessing and card classification pipeline to improve detection accuracy and eliminate manual scoring errors.
- **Technologies:** Python, OpenCV, YOLO, Kaggle Datasets, Dataset Annotation Tools, GitHub

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Multiplayer Rummy Playing Platform

- Designed and developed a full-stack Rummy playing web application enabling users to join tables and play real-time card games online.
- Implemented game logic, player turn management, and score calculation supporting standard 13-card Rummy gameplay rules.
- Developed secure backend APIs with authentication and role-based access control to manage players, game sessions, and transactions.
- **Technologies:** React.js, Node.js, Express.js, MongoDB/PostgreSQL, JWT Authentication, Railway Deployment, GitHub

Distributed Microservices E-Commerce System

- Developedascalable e-commerce backendusingmicroservices architecture with services for User, Product, Order, Payment, and Notification management.
- Implemented ASP.NET Core microservices with message queues (RabbitMQ) and optimized performance using Redis caching and multi-database support (SQL Server, MongoDB).
- **Technologies:** ASP.NET Core, RabbitMQ, SQL Server, MongoDB, Redis, Docker, Kubernetes, Ocelot.

Full-Stack Social Media Super App

- Engineereda full-stack socialmedia platform combining messaging, media sharing, and ephemeral content features using React.js, Node.js, and Express.js.
- Implemented real-time communication using Socket.io supporting chat, live location sharing, and dynamic content feeds, demonstrating scalable backend and system design principles.
- **Technologies:** React.js, Node.js, Express.js, Socket.io, PostgreSQL/MySQL, MongoDB, REST APIs, Git.

EDUCATION

Bachelor of Technology (B.Tech) – Electronics and Communication Engineering

- SreenidhiInstitute of ScienceandTechnology(SNIST),Hyderabad
- CGPA: 8.85 / 10 (up to 4th semester)
- Expected Graduation: 2027

Intermediate (Class XII)

- Bhashyam Junior College, Guntur
- Score: 96.2%

Secondary School (Class X)

- Dreamz PrimarySchool,Pusapadu
- Score: 100%

ACHIEVEMENTS & RECOGNITION

Published Patent

- Autonomous Artificial Intelligence Driven Multi-Modal Surveillance System for Precision Agricultural Monitoring - Published patent based on the Smart Agricultural Charging Line-Following Bot project.
- Proposed an AI-driven autonomous monitoring system integrating robotics, environmental sensors, IoT connectivity, and wireless inductive charging for precision agriculture.
- **Details:** Patent Application No: 202541129931 | Published Jan 2026

Research Publication

- “FatiguePreventionSystem”, published in the International Journal of Engineering Research & Technology (IJERT).
- Proposed an automotive driver fatigue detection system using camera-based monitoring and physiological sensors to identify driver drowsiness and trigger real-time alerts for accident prevention.
- **Link:** https://www.ijert.org/fatigue-prevention-system

Internship - Wireless Communication Systems

- Completeda3-weekinternshipat KramatiITSolutions LLP in the domain of Massive MIMO, mmWave, and Terahertz technologies for 5G and 6G communication systems.
- Gained exposure to next-generation wireless communication architectures, antenna technologies, and high-frequency communication systems.
- **Details:** Kramati IT Solutions LLP | Oct-Nov 2025